

Package ‘robFJM’

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Type Package

Title Robust Functional Joint Models

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Description Functions for fitting robust functional joint models for longitudinal scalar responses, time-to-event outcomes, and functional predictors. The package implements the proposed two-stage robust estimation framework based on robust functional principal component analysis and Student-t adjusted profile h-likelihood estimation. Functional predictors entering the longitudinal and survival submodels are first represented by robust functional principal component scores, reducing the influence of atypical curves and functional artifacts. Conditional on these robust scores, the joint model is fitted through an adjusted profile h-likelihood approach with Student-t longitudinal errors, observation-specific downweighting, Weibull baseline hazard, shared random effects, numerical quadrature for survival integrals, and reconstruction of the longitudinal and survival functional coefficient functions. The package also includes simulation tools for generating clean and contaminated functional joint model data.

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add_scores

*Add Robust Functional Principal Component Scores***Description**

Adds robust functional principal component score representations to the longitudinal and survival data sets used in the proposed robust functional joint modeling framework.

Usage

```
add_scores(long, surv, grid_cols_X, grid_cols_W,
           gp, rf pca_nbasis = 15, rf pca_npc_max = NULL)
```

Arguments

long	A data frame containing the longitudinal observations. The data frame must include a subject identifier column named ID, together with the functional predictor columns specified by grid_cols_X. In the proposed functional joint model, these functional measurements enter the longitudinal submodel.
surv	A data frame containing the subject-level survival information. The data frame must include a subject identifier column named ID, together with the functional predictor columns specified by grid_cols_W. In the proposed functional joint model, these functional measurements enter the survival submodel.
grid_cols_X	A character vector or integer vector identifying the columns of long that contain the longitudinal-side functional predictor values observed on the common grid gp.
grid_cols_W	A character vector or integer vector identifying the columns of surv that contain the survival-side functional predictor values observed on the common grid gp.
gp	A numeric vector giving the grid points at which the functional predictors are observed. The length of gp must match the number of functional grid columns selected by grid_cols_X and grid_cols_W.
rfpca_nbasis	An integer specifying the number of B-spline basis functions used in the robust functional principal component analysis. The default is 15.
rfpca_npc_max	An optional integer giving the maximum number of robust functional principal components to retain. If NULL, the number of retained components is chosen automatically by the robust FPCA step.

Details

The function is a preprocessing wrapper for the proposed robust functional joint model. It extracts the functional predictor matrix from the longitudinal data and the functional predictor matrix from the survival data, and then applies robust functional principal component analysis separately to the two functional predictors.

Let $\mathcal{X}_i(s)$ denote the functional predictor entering the longitudinal submodel and $\mathcal{W}_i(s)$ denote the functional predictor entering the survival submodel. The proposed method represents the functional linear terms

$$\int \mathcal{X}_i(s)\beta(s) ds \quad \text{and} \quad \int \mathcal{W}_i(s)\gamma(s) ds$$

through low-dimensional robust score vectors. Specifically, robust FPCA is used to obtain estimated eigenfunctions $\hat{\phi}_1^{(\mathcal{X},R)}, \dots, \hat{\phi}_{K_{\mathcal{X}}}^{(\mathcal{X},R)}$ and $\hat{\phi}_1^{(\mathcal{W},R)}, \dots, \hat{\phi}_{K_{\mathcal{W}}}^{(\mathcal{W},R)}$. The corresponding robust scores are then attached to the data frames as list-columns named `FPCscore.X` and `FPCscore.W`.

In contrast to classical FPCA, the robust FPCA step is designed to reduce the effect of atypical curves and local functional artifacts on the extracted score representation. These robust scores are then used as finite-dimensional predictors in `fit_FAPHL`, where the joint model is estimated by the proposed Student-*t* adjusted profile *h*-likelihood method.

The current implementation returns a nested list with one top-level element, `robust`. Therefore, downstream calls should use objects such as `out$robust$long`, `out$robust$surv`, `out$robust$FPCA.X`, and `out$robust$FPCA.W`.

Value

A list with the following component:

robust A list containing the robust FPCA-transformed data and fitted robust FPCA objects:

long A copy of `long` with an added list-column `FPCscore.X` containing the robust FPCA scores for the longitudinal-side functional predictor.

surv A copy of `surv` with an added list-column `FPCscore.W` containing the robust FPCA scores for the survival-side functional predictor.

FPCA.X The robust FPCA object for the longitudinal-side functional predictor. This object contains the robust scores, retained basis evaluations, number of retained components, and additional robust FPCA output.

FPCA.W The robust FPCA object for the survival-side functional predictor. This object contains the robust scores, retained basis evaluations, number of retained components, and additional robust FPCA output.

Note

This function is intended to be used before `fit_FAPHL`. The input data should be organized so that each subject has a unique ID; the longitudinal data may contain repeated rows per subject, whereas the survival data should contain one row per subject.

See Also

[process_rfpca](#), [fit_FAPHL](#), [simulate_data](#)

Examples

```
I <- 100; J <- 4; D <- 10; by <- 0.1

# Contamination settings
outlier_frac_Y <- 0.05
outlier_frac_fun <- 0.05

dat <- simulate_data(
  I = I, J = J, D = D, by = by,
  outlier_frac_Y = outlier_frac_Y,
  outlier_frac_fun = outlier_frac_fun
)
```

```

sgrid <- seq(0, D, by)
m <- length(sgrid)
grid_cols_X <- paste0("func.X.", 1:m)
grid_cols_W <- paste0("func.W.", 1:m)

sc <- add_scores(
  long = dat$data.long,
  surv = dat$data.surv,
  grid_cols_X = grid_cols_X,
  grid_cols_W = grid_cols_W,
  gp = dat$grid,
  rfpc_nbasis = 10
)

names(sc)
names(sc$robust)
dim(sc$robust$FPCA.X$FPCscore)
dim(sc$robust$FPCA.W$FPCscore)

```

fit_FAPHL

Fit a Robust Functional Joint Model

Description

Fits the proposed robust functional joint model using robust FPCA scores and Student- t adjusted profile h -likelihood estimation.

Usage

```
fit_FAPHL(long, surv, FPCA.X, FPCA.W, by,
          nu = 4, verbose = FALSE)
```

Arguments

long	A longitudinal data frame containing repeated scalar responses and robust FPCA scores for the longitudinal-side functional predictor. The data frame must include the columns ID, Y, obstime, and FPCscore.X. The column ID is the subject identifier, Y is the scalar longitudinal response, obstime is the longitudinal observation time, and FPCscore.X is the robust FPCA score representation of the functional predictor entering the longitudinal submodel.
surv	A subject-level survival data frame. The data frame must include the columns ID, time, event, W1, and FPCscore.W. The column time is the observed event or censoring time, event is the event indicator, W1 is a scalar survival-side explanatory variable, and FPCscore.W is the robust FPCA score representation of the functional predictor entering the survival submodel.
FPCA.X	The robust FPCA object for the longitudinal-side functional predictor. This is typically obtained from add_scores and should contain the robust basis evaluations in FPCA.X\$basis.
FPCA.W	The robust FPCA object for the survival-side functional predictor. This is typically obtained from add_scores and should contain the robust basis evaluations in FPCA.W\$basis.

by	A numeric value giving the spacing of the functional grid. This argument is kept for consistency with the simulation and score-construction interface. The coefficient functions are reconstructed on the grid represented in <code>FPCA.X\$basis</code> and <code>FPCA.W\$basis</code> .
nu	Degrees of freedom parameter for the Student- t longitudinal error distribution. Smaller values yield heavier tails and stronger downweighting of large longitudinal residuals. The default is 4.
verbose	Logical argument reserved for optional progress output. The default is FALSE.

Details

The function implements the main estimation step of the proposed robust functional joint model. It assumes that robust functional dimension reduction has already been performed, usually by [add_scores](#). The longitudinal and survival functional predictors are therefore represented by robust FPCA scores, which enter the two model components as finite-dimensional explanatory variables.

The fitted longitudinal submodel has the working form

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + \boldsymbol{\xi}_i^{(\mathcal{X})\top} \boldsymbol{\beta}_x + u_i + \varepsilon_{ij},$$

where Y_{ij} is the scalar longitudinal response, t_{ij} is the observation time, $\boldsymbol{\xi}_i^{(\mathcal{X})}$ is the robust FPCA score vector for the longitudinal-side functional predictor, and u_i is a shared subject-specific random effect.

The survival submodel uses a Weibull proportional hazards structure with a current-value association term,

$$h_i(t) = \tau t^{\tau-1} \exp\{\gamma_1 W_{1i} + \boldsymbol{\xi}_i^{(\mathcal{W})\top} \boldsymbol{\gamma}_w + \alpha m_i(t)\},$$

where $\boldsymbol{\xi}_i^{(\mathcal{W})}$ is the robust FPCA score vector for the survival-side functional predictor, $m_i(t)$ is the latent longitudinal trajectory, α is the association parameter, and τ is the Weibull shape parameter.

Robustness to aberrant longitudinal observations is introduced through a Student- t error distribution. Using the Gaussian scale-mixture representation,

$$\varepsilon_{ij} \mid \lambda_{ij} \sim N(0, \sigma_\varepsilon^2 / \lambda_{ij}), \quad \lambda_{ij} \sim \text{Gamma}(\nu/2, \nu/2),$$

the method updates observation-specific weights according to

$$\hat{\lambda}_{ij} = \frac{\nu + 1}{\nu + r_{ij}^2 / \sigma_\varepsilon^2},$$

where r_{ij} is the current residual. Large residuals therefore receive smaller weights.

The function then calls the internal routine `fit_APHL`, which maximizes an adjusted profile h -likelihood. The random effects are profiled by damped Newton updates, survival integrals are evaluated numerically, and the Laplace-type curvature correction is used to adjust the profiled h -likelihood. After estimating the score-level coefficients, the function maps them back to the functional domain to obtain $\hat{\beta}(s)$ and $\hat{\gamma}(s)$.

Value

A list with the following components:

fit	A list returned by the internal adjusted profile h -likelihood fitting routine. It contains the estimated standardized-model parameters after back-transformation, including fixed-effect vectors, Weibull shape parameter, association parameter, variance components, and estimated random effects.
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BetaHat	A numeric vector containing the reconstructed estimate of the functional coefficient $\beta(s)$ for the longitudinal-side functional predictor, evaluated on the grid represented by <code>FPCA.X\$basis</code> .
GammaHat	A numeric vector containing the reconstructed estimate of the functional coefficient $\gamma(s)$ for the survival-side functional predictor, evaluated on the grid represented by <code>FPCA.W\$basis</code> .
scalars	A named numeric vector containing the main scalar parameter estimates: <code>beta1</code> , <code>var_b</code> , <code>var_eps</code> , <code>gamma1</code> , <code>alpha</code> , and <code>Tau</code> . Here <code>beta1</code> is the longitudinal time effect, <code>var_b</code> is the random-intercept variance estimate, <code>var_eps</code> is the longitudinal error variance estimate, <code>gamma1</code> is the scalar survival-side coefficient for <code>W1</code> , <code>alpha</code> is the current-value association parameter, and <code>Tau</code> is the Weibull shape parameter.

Note

This function expects the robust FPCA score columns to be present in the input data. The recommended workflow is to first call [simulate_data](#) or prepare user data in the required format, then call [add_scores](#), and finally fit the proposed model with `fit_FAPHL`.

The current implementation uses a random-intercept shared-effects structure and a Weibull baseline hazard. The scalar survival explanatory variable is expected to be named `W1`.

See Also

[add_scores](#), [process_rfpcas](#), [simulate_data](#)

Examples

```
I <- 100; J <- 4; D <- 10; by <- 0.1

# Contamination settings
outlier_frac_Y <- 0.05
outlier_frac_fun <- 0.05

dat <- simulate_data(
  I = I, J = J, D = D, by = by,
  outlier_frac_Y = outlier_frac_Y,
  outlier_frac_fun = outlier_frac_fun
)

sgrid <- seq(0, D, by)
m <- length(sgrid)
grid_cols_X <- paste0("func.X.", 1:m)
grid_cols_W <- paste0("func.W.", 1:m)

sc <- add_scores(
  long = dat$data.long,
  surv = dat$data.surv,
  grid_cols_X = grid_cols_X,
  grid_cols_W = grid_cols_W,
  gp = dat$grid,
  rfpcas_nbasis = 10
)
```

```

fit <- fit_FAPHL(
  long = sc$robust$long,
  surv = sc$robust$surv,
  FPCA.X = sc$robust$FPCA.X,
  FPCA.W = sc$robust$FPCA.W,
  by = by, verbose = FALSE
)

fit$scalars
length(fit$BetaHat)
length(fit$GammaHat)

plot(dat$grid, fit$BetaHat, type = "l",
      xlab = "Functional domain",
      ylab = "Estimated beta(s)")

plot(dat$grid, fit$GammaHat, type = "l",
      xlab = "Functional domain",
      ylab = "Estimated gamma(s)")

```

process_rfpca

*Process Functional Predictors by Robust FPCA***Description**

Computes robust functional principal component scores and basis evaluations for functional predictors observed on a common grid.

Usage

```
process_rfpca(subjs, funcs_mat, gp, nbasis = 15, npc_max = NULL)
```

Arguments

subjs	A vector of subject identifiers. Its length must match the number of rows of funcs_mat. If repeated rows are present for the same subject, robust FPCA is computed using the first occurrence of each subject and the resulting scores are mapped back to all rows according to subjs.
funcs_mat	A numeric matrix containing the observed functional predictor values. Rows correspond to observations or subjects and columns correspond to grid points. The number of columns must match the length of gp.
gp	A numeric vector giving the grid points on which the functional predictor is observed.
nbasis	An integer specifying the number of B-spline basis functions used to smooth and represent the functional observations before robust FPCA. The default is 15.
npc_max	An optional integer specifying the maximum number of robust functional principal components to retain. If NULL, the number of retained components is selected automatically by the robust FPCA step.

Details

This function is the core robust functional dimension-reduction routine used by the proposed robust functional joint modeling procedure. It first identifies the subject-level functional observations by retaining the first occurrence of each subject identifier in `subjs`. Robust FPCA is then applied to these baseline subject-level functional observations through the internal routine `getRPCA`.

The robust FPCA step represents each function in a B-spline basis and computes robust principal directions using a projection-pursuit covariance estimator. Let $\mathcal{X}_i(s)$ denote a functional predictor observed on the grid `gp`. The method approximates

$$\mathcal{X}_i(s) \approx \sum_{\ell=1}^{K_b} c_{i\ell} \psi_{\ell}(s),$$

where $\{\psi_{\ell}\}_{\ell=1}^{K_b}$ is a B-spline basis. A robust center is used for the spline coefficient vectors, and robust principal directions are then extracted from a projection-pursuit covariance estimate. This approach is designed to reduce the influence of atypical curves and local functional artifacts on the estimated eigenfunctions.

If `npc_max` = NULL, the function uses the number of robust components selected by `getRPCA`. If `npc_max` is supplied, the number of retained components is

$$K = \min(K_{\text{rob}}, \text{npc_max}),$$

where K_{rob} is the number of components selected by the robust FPCA step. At least one component is retained.

The output score matrix `FPCscore` has the same number of rows as `funcs_mat`. Therefore, when the input contains repeated longitudinal records for the same subject, each repeated record receives the same subject-level robust FPCA score vector. This convention is useful in the functional joint model because the functional predictor is treated as a baseline subject-level predictor, while the longitudinal scalar response may be measured repeatedly over time.

The returned basis evaluations in `basis` are evaluated on `gp` and are used by `fit_FAPHL` to reconstruct the estimated functional coefficient functions after score-level model fitting.

Value

A list with the following components:

<code>FPCscore</code>	A numeric matrix of robust functional principal component scores. The number of rows equals the number of rows of <code>funcs_mat</code> , and the number of columns equals the number of retained robust components. Columns are named PC1, PC2, and so on.
<code>kz</code>	The number of retained robust functional principal components.
<code>basis</code>	A numeric matrix containing the retained robust eigenfunctions evaluated on the grid <code>gp</code> . Rows correspond to grid points and columns correspond to retained components.
<code>rp</code>	The full robust FPCA object returned by <code>getRPCA</code> . This object contains additional information such as the functional principal component object, robust scores, B-spline basis, and grid information.

Note

This function is usually not called directly by end users. The recommended workflow is to call [add_scores](#), which applies `process_rfpca` separately to the longitudinal-side and survival-side functional predictors and attaches the resulting robust scores to the data.

The function assumes that the functional predictor is observed on a common grid for all subjects. Missing values in `funcs_mat` should be handled before calling this function.

See Also

[add_scores](#), [fit_FAPHL](#), [simulate_data](#)

Examples

```
I <- 100; J <- 4; D <- 10; by <- 0.1

# Contamination settings
outlier_frac_Y <- 0.05
outlier_frac_fun <- 0.05

dat <- simulate_data(
  I = I, J = J, D = D, by = by,
  outlier_frac_Y = outlier_frac_Y,
  outlier_frac_fun = outlier_frac_fun
)

sgrid <- seq(0, D, by)
m <- length(sgrid)
grid_cols_X <- paste0("func.X.", 1:m)

Xmat <- as.matrix(dat$data.long[, grid_cols_X, drop = FALSE])

rfpca_x <- process_rfpca(
  subjs = dat$data.long$ID,
  funcs_mat = Xmat,
  gp = dat$grid,
  nbasis = 10
)

names(rfpca_x)
dim(rfpca_x$FPCscore)
rfpca_x$kz

matplot(dat$grid, rfpca_x$basis, type = "l",
  xlab = "Functional domain",
  ylab = "Robust eigenfunctions")
```

Description

Generates synthetic longitudinal, survival, and functional predictor data for illustrating and evaluating the proposed robust functional joint model.

Usage

```
simulate_data(I = 200, J = 4,
             D = 10, by = 0.1,
             beta0 = 1.35, beta1 = 0.78,
             var_b = 1, var_eps = 1,
             gamma0 = 0, gamma1 = -1.75, alpha = 0.29,
             Tau = 0.8,
             lambdaC = -6.0,
             Kx_latent = 5, Kw_latent = 5,
             meas_sd_X = 0.15, meas_sd_W = 0.15,
             outlier_frac_Y = 0.00, outlier_mag_Y = c(20, 30),
             outlier_frac_fun = 0.00, outlier_mag_fun = c(3, 6))
```

Arguments

I	An integer specifying the number of subjects to generate. The default is 200.
J	An integer specifying the planned number of repeated longitudinal measurements per subject before trimming measurements after the observed follow-up time. The default is 4.
D	A numeric value giving the upper limit of the functional domain and the planned longitudinal follow-up window. The functional grid is defined on $[0, D]$. The default is 10.
by	A numeric value giving the spacing between adjacent functional grid points. The grid is generated as <code>seq(0, D, by)</code> . The default is 0.1.
beta0	The intercept parameter in the longitudinal submodel. The default is 1.35.
beta1	The longitudinal time-effect parameter. The default is 0.78.
var_b	The variance of the subject-specific random intercept in the longitudinal submodel. The default is 1.
var_eps	The variance of the scalar longitudinal measurement error. The default is 1.
gamma0	The intercept term used in the survival linear predictor. The default is 0.
gamma1	The scalar survival-side coefficient for the binary explanatory variable W_1 . The default is -1.75.
alpha	The current-value association parameter linking the latent longitudinal trajectory to the hazard function. The default is 0.29.
Tau	The Weibull shape parameter used in both event-time and censoring-time generation. The default is 0.8.
lambdaC	A log-scale censoring parameter controlling the censoring-time distribution. Larger values generally produce shorter censoring times and hence more censoring. The default is -6.0.
Kx_latent	The number of latent basis functions used to generate the functional predictor entering the longitudinal submodel. The default is 5.
Kw_latent	The number of latent basis functions used to generate the functional predictor entering the survival submodel. The default is 5.

meas_sd_X	The standard deviation of the measurement noise added to the longitudinal-side functional predictor. The default is 0.15.
meas_sd_W	The standard deviation of the measurement noise added to the survival-side functional predictor. The default is 0.15.
outlier_frac_Y	The proportion of scalar longitudinal observations contaminated by additive outliers. The default is 0.00, corresponding to no scalar-response contamination.
outlier_mag_Y	A numeric vector of length two giving the lower and upper bounds for the absolute magnitude of scalar longitudinal outliers. The sign is randomly assigned. The default is c(20, 30).
outlier_frac_fun	The proportion of subjects whose functional predictors are contaminated by subject-level magnitude shifts. The default is 0.00, corresponding to no functional contamination.
outlier_mag_fun	A numeric vector of length two giving the lower and upper bounds for the absolute magnitude of functional-predictor shifts. The sign is randomly assigned. The default is c(3, 6).

Details

This function generates data from a functional joint model with a scalar longitudinal response, a time-to-event outcome, and two baseline functional predictors. One functional predictor enters the longitudinal submodel and the other enters the survival submodel.

The longitudinal submodel is generated according to

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + \int_0^D X_i(s) \beta(s) ds + b_i + \varepsilon_{ij},$$

where $b_i \sim N(0, \sigma_b^2)$ is a subject-specific random intercept and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ is the scalar longitudinal error. In the implementation, σ_b^2 is controlled by `var_b` and σ_ε^2 is controlled by `var_eps`.

The survival submodel uses a Weibull current-value joint modeling structure. The subject-specific hazard is generated from

$$h_i(t) = \tau t^{\tau-1} \exp\{\gamma_0 + \gamma_1 W_{1i} + \int_0^D W_i(s) \gamma(s) ds + \alpha m_i(t)\},$$

where $m_i(t)$ is the latent longitudinal trajectory. Event times are generated by numerical inversion of the cumulative hazard. Censoring times are generated independently from a Weibull-type distribution controlled by `lambdaC` and `Tau`.

The true coefficient functions are

$$\beta(s) = 2 \sin(\pi s/5), \quad \gamma(s) = 1.2 \sin(\pi s/4).$$

The latent functional predictors are generated using finite sine and cosine basis expansions with decreasing score variances. Measurement noise is then added to obtain the observed functional predictors.

Two contamination mechanisms can be introduced. First, scalar longitudinal responses can be contaminated by additive outliers using `outlier_frac_Y` and `outlier_mag_Y`. Second, functional predictors can be contaminated by subject-level vertical shifts using `outlier_frac_fun` and `outlier_mag_fun`. These mechanisms are useful for evaluating the robustness of the proposed robust FPCA plus Student-*t* adjusted profile *h*-likelihood estimator.

After generating the full longitudinal data, observations occurring after the observed event or censoring time are removed. Thus, the returned longitudinal data mimic the usual joint-modeling setting in which repeated measurements are only available up to follow-up.

Value

A list with the following components:

data.long A longitudinal data frame containing repeated measurements. It includes subject identifiers ID, visit numbers `visit`, observation times `obstime`, observed follow-up times `time`, event indicators `event`, scalar longitudinal responses `Y`, functional predictor columns named `func.X.1`, `func.X.2`, and so on, and the generated random intercept `b`.

data.surv A subject-level survival data frame containing ID, observed follow-up time `time`, event indicator `event`, scalar explanatory variable `W1`, and functional predictor columns named `func.W.1`, `func.W.2`, and so on.

grid The numeric grid on which the functional predictors and true coefficient functions are evaluated.

by The functional grid spacing used to generate `grid`.

true A list containing the true parameter values and true coefficient functions used in the data-generating process. It includes `beta0`, `beta1`, `var_b`, `var_eps`, `gamma0`, `gamma1`, `alpha`, `Tau`, `trueBeta`, and `trueGamma`.

See Also

[add_scores](#), [process_rfpca](#), [fit_FAPHL](#)

Examples

```
I <- 100; J <- 4; D <- 10; by <- 0.1

# Contamination settings
outlier_frac_Y <- 0.05
outlier_frac_fun <- 0.05

dat <- simulate_data(
  I = I, J = J, D = D, by = by,
  outlier_frac_Y = outlier_frac_Y,
  outlier_frac_fun = outlier_frac_fun
)

names(dat)
head(dat$data.long)
head(dat$data.surv)
```

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